Neural Control & Co-ordination - 100 MCQs

NCERT Nichod : Biology Practice Series NEET 2025

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10AM

Good evening

audible?

Ready

Neural Control and Coordination

BIOLOGYLIVE

1. The cerebral cortex is referred to as :-

(1) White matter-axon

(2) White matter- cell bodies

(3) Grey matter- axon

(4) Grey matter- cell bodies

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2. The inner part of cerebral hemisphere & group of associated deep structure form a complex structure called limbic system. The associated deep structures include:-

i. Amygdala

ii. Hippocampus

iii. Outer parts of cerebral cortex

iv. Association area

(1) i & iv (2) i & ii

(3) i & iii (4) iii & iv

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3. "Axons" are different from the cyton is

(1) Presence of nissl's granules

(2) Absence of nissl's granules → Axon do not have Niss's

(3) Presence of schwann cells always

(4) Presence of nodes of ranvier always

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4. Assertion : Transmission of a nerve impulse across a synapse is brought by a neurotransmitter.

Reason : A neurotransmitter is necessary to transmit a nerve impulse across a synapse because there is a small gap, the synaptic cleft between the two neurons at the synapse.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.

5. Corpus callosum :- tract of Nerve Fibres

(1) Is a tract of connective tissue fibres

(2) Is a tract of dense muscular fibres

(3) Is a tract of dense irregular connective tissue fibres

(4) None is correct

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6. Myelinated nerve fibre is found in

(1) Spinal nerve

(2) Cranial nerve

(3) Both (1) and (2)

(4) None of these

7. The parts of human brain that helps in regulation of sexual behaviour, expression of excitement, pleasure, rage, fear etc. are:

(1) Limbic system and hypothalamus

(2) Corpora quadrigemina and hippocampus

(3) Brain stem and epithalamus

(4) Corpus callosum and thalamus

8. Select the correct answer :-

i. The sensory organs detect all types of changes in the environment and send appropriate signals to the CNS.

ii. Signals are then sent to different parts/ centres of the brain.

(1) i is correct

(2) ii is correct

(3) Both are correct

(4) Both are incorrect

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9. Arrange the following events of synaptic transmission in a proper sequence.

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A. Binding of neurotransmitter to specific receptors on the postsynaptic membrane. 2

B. Generation of a new potential in the postsynaptic neuron. 4

C. Release of neurotransmitter into the synaptic cleft. → 1

D. Opening of ion channels in the postsynaptic membrane. 3

(1) A, C, D, B (2) C, A, B, D

(3) C, A, D, B ✓ (4) D, A, C, B

10. Synaptic knob is bulb-like structure which is present

(1) At the end of axon terminal

(2) At the node of Ranvier

(3) At the end of cytonic terminal

(4) At the end of dendrites

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11. Statement A : Myelinated nerve fibres are found in autonomous and the somatic neural systems. X

Statement B : Unmyelinated nerve fibres are found in spinal and cranial nerves. X

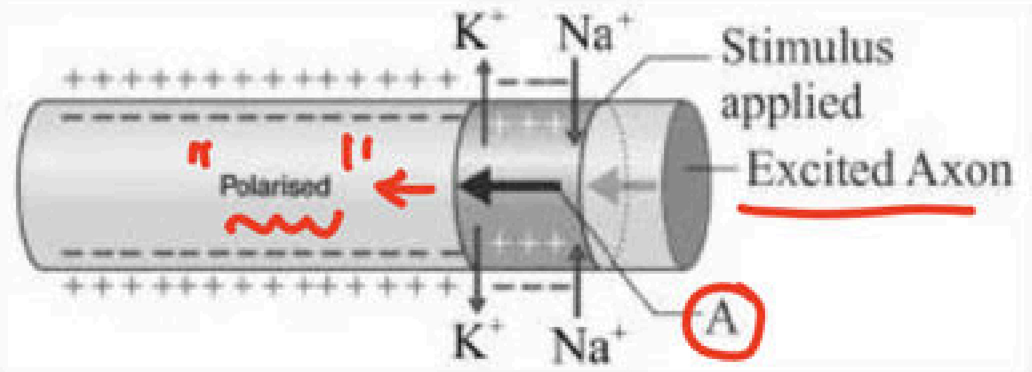
(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

12. Following is the diagrammatic representation of impulse conduction through an axon. Identify the process of impulse conduction in ?



(1) Action potential

(2) Resting potential

(3) Deporalisation

(4) None of above

13. Brain stem is formed by

(1) Fore brain

(2) Mid brain

(3) Pons and medulla oblongata

(4) 2 and 3

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14. If the receptors are "removed" from postsynaptic membrane, then :-

(1) Synaptic transmission will be faster

(2) Chemical synaptic transmission will become slow

(3) Chemical synaptic transmission will not occur

(4) Synaptic transmission will not be affected

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15. The function of the organs/organ systems in our body must be coordinated to maintain :-

(1) Neural coordination

(2) Homeostasis

(3) Respiration

(4) All are correct

16. Read the given statements and select the correct option .

(1) The thalamus wraps around a structure called cerebrum, which is major coordinating centre for sensory and motor signaling

(2) Corpus callosum is referred to as grey matter due to it's greyish appearance

middle

(3) Arachnoid is the outer most layer of the meninges

(4) Hypothalamus lies at the floor of the thalamus

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17. Statement A : ~~PNS~~ CNS processes and analyses all the sensory inputs

Statement B : Receptors receive ~~motor impulses~~ Stimulus ×

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

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18. Which of the following has a very convoluted surface in order to provide additional space for many neurons?

(1) Thalamus

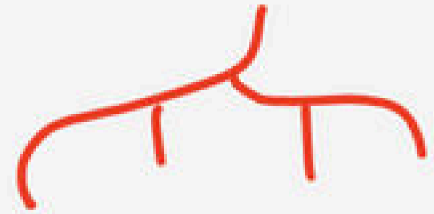
(2) Pons

(3) Cerebellum

(4) Hypothalamus

19. Statement A : Short fibres which branch repeatedly and project out of the "cell body" also contain Nissl's granules and called dendrites.

Statement B : Axon is a long fibre, the distal end of which is branched.



(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

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20. Statement A : The neural system provides an organized network of point - to - point connections for a quick coordination.

Statement B : The endocrine system provides chemical integration through hormones.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

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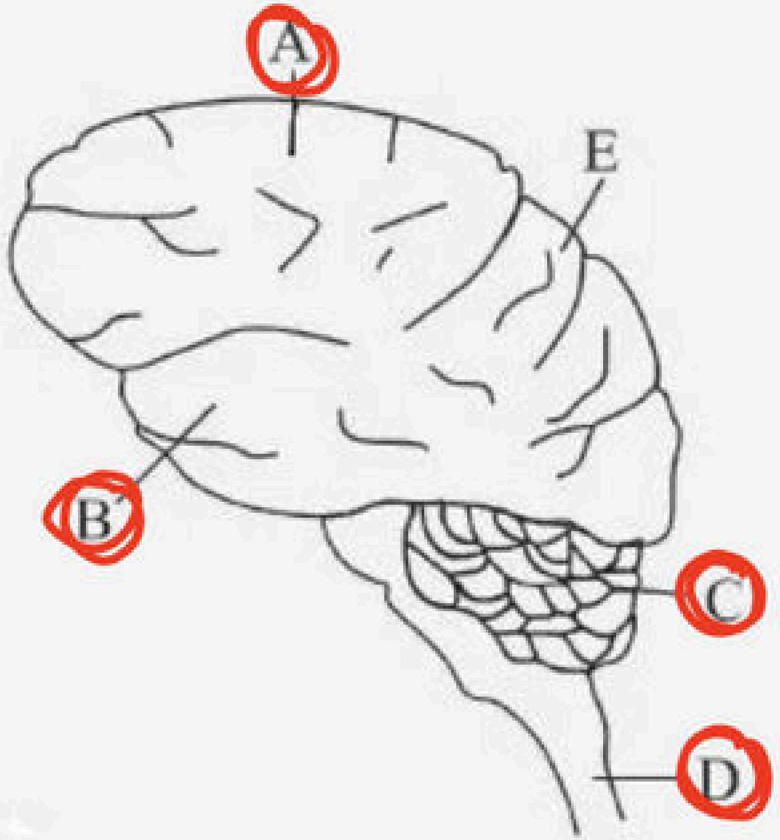
21. In the diagram of the lateral view of the human brain, parts are indicated by alphabets. Choose the answer in which those alphabets have been correctly matched with the parts which they indicate



(1) A - Temporal lobe; B - Parietal lobe; C - Cerebrum; D - medulla oblongata; E - Frontal lobe

(2) A - Frontal lobe; B - Temporal lobe; C - Cerebellum; D - Medulla oblongata; E - Parietal lobe

(3) A - Temporal lobe; B - Paretial lobe; C - Cerebellum; D - Medulla oblongata; E - Frontal lobe

(4) A - Frontal lobe; B - Temporal lobe; C - Cerebrum; D - Medulla oblongata; E - Occipital lobe

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22. In reflex action the entire process of response is to a ___.

(1) CNS

(2) Spinal cord

(3) Peripheral Nervous Stimulation

(4) Voluntary response

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23. Select the incorrect statement:

(1) In Hydra endocrinal organisation is composed of a network of neurons. x

(2) The neural system is better organised in insects, where a brain is present along with a ~~single~~ number ganglia and neural tissues. x

(3) The neural system of all animals is composed of highly specialised cells called nephrons x

(4) All are incorrect.

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24. The synaptic vesicle moves towards the membrane, where it fuses with plasma membrane and release neurotransmitter. The membrane from where the neurotransmitter released is ___

(1) Special membrane of the axon

(2) Pre-synaptic membrane

(3) Synaptic cleft membrane

(4) Post-synaptic membrane

25. Which of the following consists of fibre tracts interconnecting the different regions of brain?

(1) Cerebellum

(2) Pons varolli

(3) Medulla

(4) Spinal cord

Outside

26. White matter is Inside in brain and____ in spinal cord in V.V.Imp

humans

(1) Outside, inside

(2) Inside, Outside

(3) Inside, Inside

(4) Outside, Outside

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27. Nervous system is different from endocrine because it shows :-

V.V.Imp

i. Point to point effect

ii. Widely spread effect x

iii. Slow speed of service x

iv. Fast speed of service

(1) i and iv

(2) i and iii

(3) ii and iii

(4) iii and iv

28. The most abundant intracellular cation is

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(1) H^+

(2) K^+

(3) Na^+

(4) Ca^{++}

Title: Neural Control and Coordination

29. Assertion : Medulla contains centres which control respiration, cardiovascular reflexes and gastric secretions. → Nuclie

Reason : ~~Medulla~~ contains several neurosecretory cells which secrete hormones. Hypoth.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.

30. A man is admitted to a hospital. He is suffering from an abnormally low body temperature, loss of appetite and extreme thirst. His brain scan would probably show a tumour in

(1) Pons

(2) Cerebellum

(3) Hypothalamus

(4) Medulla Oblongata

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31. The electrical potential difference across the plasma membrane is called the____.

Which is infact termed as a nerve impulse.

(1) Potential deficit

(2) Resting potential

(3) Action potential

(4) All of these

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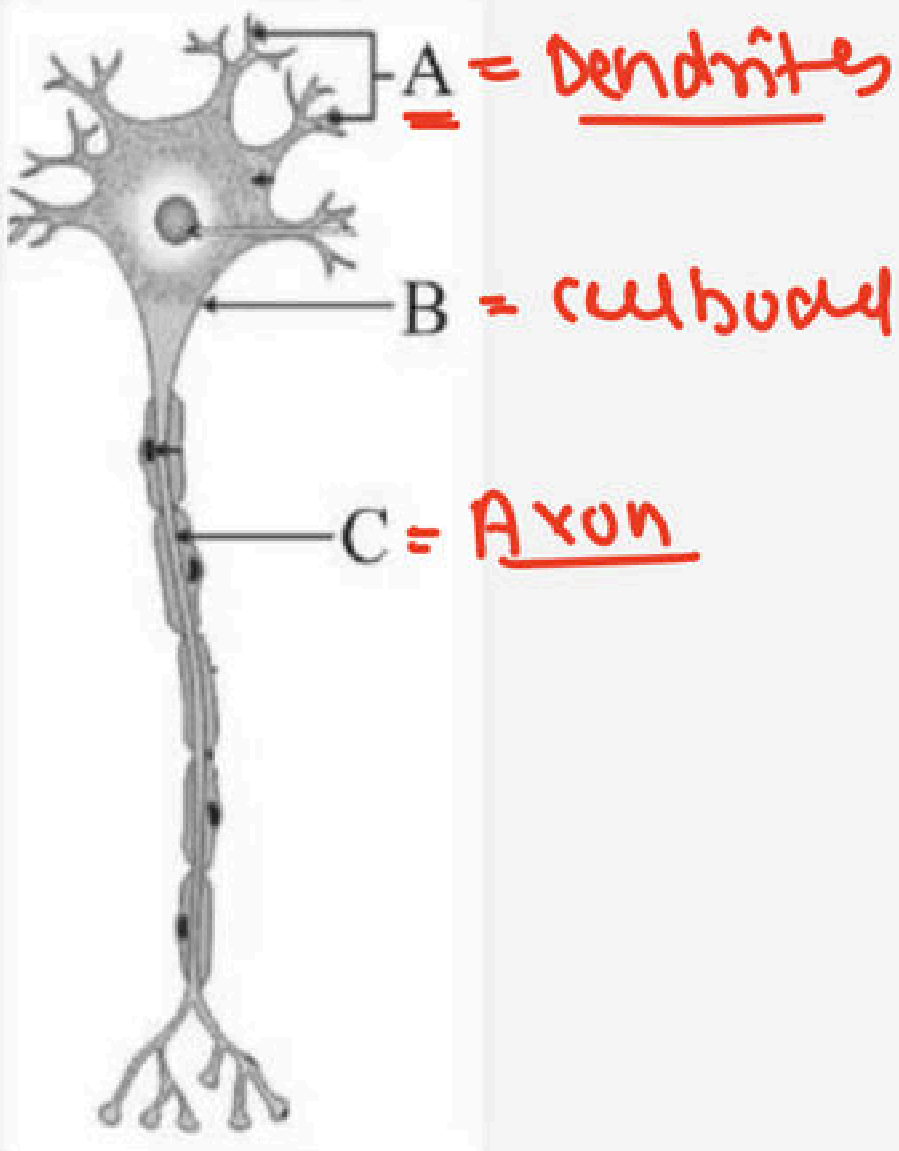
32. Identify the three major parts of neuron (A-C) in the given diagram?



(1) A - Cell body, B - Axon,C - Dendrites

(2) A - Axon, B - Cell body, C - Dendrites

(3) A - Dendrites, B - Cell body, C - Axon

(4) A - Axon, B - Dendrites, C - Cell body

33. The brain can be divided into three major parts:

(1) Left brain, midbrain, and right brain

(2) Dorsal brain, midbrain, and ventral brain

(3) Forebrain, midbrain, and hindbrain

(4) Medial brain, midbrain, and lateral brain

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34. Assertion : Myelin sheath insulates nerve fibre and prevents its ~~depolarisation~~.

Reason : Nerve impulses are conducted more rapidly in non-myelinated fibres than myelinated ones.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.

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35. Depolarization occurs due to

(1) Rapid influx of Na^+

(2) Rapid eflux of Na^+

(3) Slow influx of Na^+

(4) Slow eflux of Na^+

36. Find out the correct match :-

(1) Limbic system - regulation of sexual behaviour with hypothalamus

(2) Pons - Consists fibre tracts that inter connect different brain regions

(3) Association-Intersensory area association, memory and communication

(4) All are correct

medulla oblong.

37. A special neural centre in the ____ can moderate the cardiac function through ____ ANS

(1) Medulla oblongata, CNS

(2) Medulla oblongata, ANS

(3) Mid brain, PNS

(4) Mid brain, medulla oblongata

38. Which of the following statements are correct in relation to Hind brain?

I. Consists of pons, cerebellum, medulla oblongata

II. Pons consists of fibre tracts that interconnect different regions of the brain

III. Cerebellum has visual area ×

IV. Medulla oblongata connected to spinal cord

(1) I, II, IV are correct (2) I, III are correct

(3) II, III are correct (4) I, II, III are correct

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39. Statement A : At resting, the axonal membrane is comparatively more permeable to potassium ions and nearly impermeable to sodium ions.

Statement B : The ionic gradients across the resting membrane are maintained by the active transport of ions by sodium-potassium pump which transports 2 sodium ions outwards for potassium ions into the cell. X

✓ (1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

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40. Assertion : Stimulus is interpreted by the brain and not by sense organs.

Reason : Sense organs act as transducers transforming the stimulus energy.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.

41. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Central Neural system	P	Relays impulses from CNS to skeletal muscles
B	Peripheral Neural system	Q	Includes brain & spinal cord
C	Autonomic Neural system	R	Transmits impulses from CNS to involuntary organs & smooth muscles
D	Somatic Neural system	S	Comprises of all nerves associated with the CNS

(1) A-P,B-R,C-Q,D-S

(2) A-Q,B-P,C-S,D-R

(3) A-Q, B-S, C-R, D-P

(4) A-P,B-Q,C-R, D-S

42. Statement A : Each branch of the axon terminates as a bulb-like structure called synaptic knob which posses synaptic vesicles containing chemicals called neurotransmitters.

Statement B : The axons transmit nerve impulses away from the cell body to a synapse or to a neuro-muscular junction.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

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43. Select incorrect statement regarding synapses:

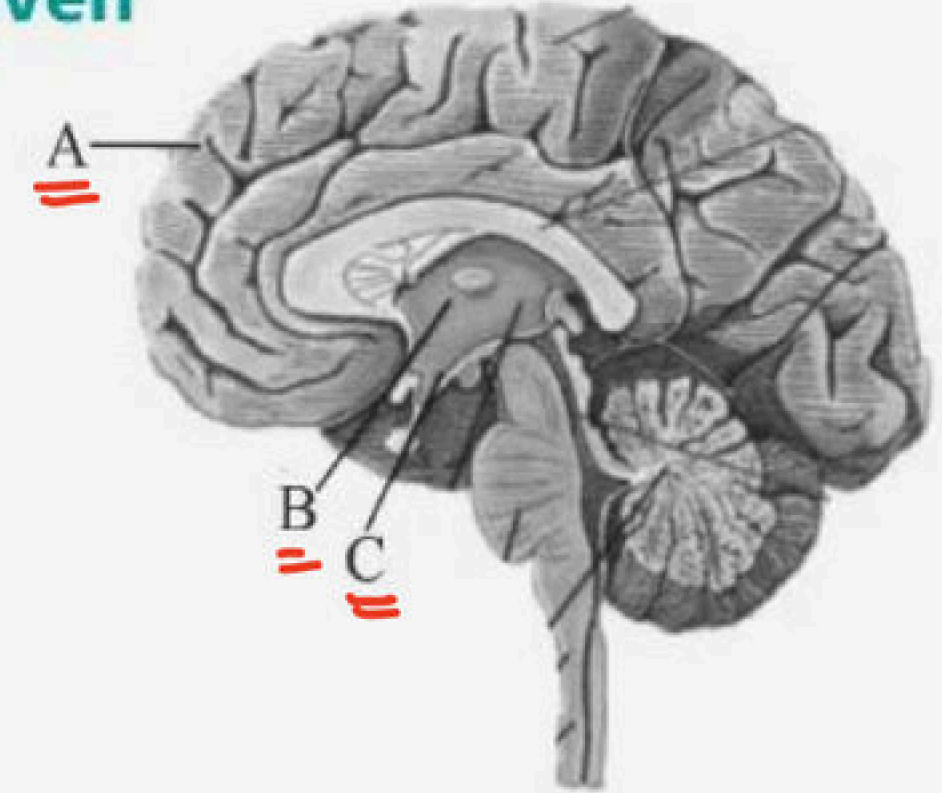
(1) Electrical current can flow directly from one neuron into the other across the electrical synapse.

(2) Chemical synapses use neurotransmitters

(3) Impulse transmission across a ~~chemical~~ electrical synapse is always faster than that across an electrical synapse.

(4) The membranes of presynaptic and postsynaptic neurons are in close proximity in an electrical synapse.

44. Identify the labeled parts of forebrain A-C in the given diagram?



(1) A - Cerebrum,B - Thalamus,C - Hypothalamus

(2) A - Thalamus, B - Hypothalamus, C - Cerebrum

(3) A - Hypothalamus, B - Thalamus, C - Cerebrum

(4) A-Cerebrum, B - Hypothalamus, C - Thalamus

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45. Match the columns I and II and choose the correct option.

Column I	Column II
A. Electrical synapse	P. Where neurotransmitter bind
B. Chemical synapse	Q. Found rare in our body
C. Receptors	R. Synaptic cleft is present

(1) A-P,B-R,C-Q

(2) A-Q, B-R, C-P

(3) A-P, B-Q,C-R

(4) A-RB-Q,C-P

46. Synaptic vesicles contains chemicals called ___ and is present in ___

(1) Synaptic fluid, axon

(2) Neurotrasmitters, cyton

(3) Neurotrasmitters, axon

(4) Neurotrasmitters, neuron

47. The inner parts of cerebral hemisphere and a group of associated deep structures like form a complex structure called :

(1) Limbs system

(2) Limbic system

(3) Corpora quadrigemina

(4) 1 and 2

48. Resting membrane potential is maintained by

(1) Hormones

(2) Neurotransmitters

(3) Ion pumps

(4) None of the above

49. Bipolar axons are found in

(1) Retina of eye

(2) Cerebral cortex

(3) Mesencephalon

(4) Embryonic stag

Title: Neural Control and Coordination

50. Statement A : The cerebrum wraps around a structure called thalamus

Statement B : The cerebral cortex is referred to as the grey matter and has ~~axon~~s. cell bodies

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

51. Identify the basic functions of neural system.

(1) Receiving sensory input from internal and external environment by nerves

(2) Processing the input information

(3) Responding to stimuli

(4) All of the above

52. Limbic system controls

(1) Sexual behaviour

(2) Motivation

(3) Affection

(4) All of them

53. Statement A : The human brain is well protected by the skull.

Statement B : Inside the skull, the brain is covered by cranial meninges consisting of an outer ~~inner~~ layer called dura mater, a very thin middle layer called arachnoid and an Inn ~~outer~~ layer called pia mater.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

54. Identify the correct pair

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(1) Multipolar - with one axon and two or more dendrites

(2) Unipolar - cell body with one axon only

(3) Bipolar- one axon and one dendrite

(4) All are correct

55. Statement A : The gap between two neurons is called synaptic cleft.

Statement B : A synapse is formed by the membranes of a pre-synaptic neuron and a post-synaptic neuron.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

✓(3) Both statements are correct.

(4) Both Statements are incorrect.

56. At ____synapses, the membranes of pre- and postsynaptic neurons are in very close proximity.

(1) Physical

(2) Electrical

(3) Chemical

(4) Biological

57. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Cell body	P	Long fiber, branched, distal end of neuron
B	Dendrites	Q	Bulb-like structure which possess synaptic vesicles
C	Synaptic knob	R	Contains cytoplasm with typical cell organelles
D	Axon	S	Short fibres which branch repeatedly and project out of the cell body

(1) A-R, B-S, C-Q, D-P
(2) A-S, B-R, C-Q, D-P
(3) A-Q, B-S, C-P, D-R
(4) A-P, B-Q, C-R, D-S

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58. Read the statements and identify which is "TRUE/FALSE"

I. The axoplasm inside the axon contains low concentration of . K^+ X

II. The axoplasm inside the axon contains high concentration of K^+ **and negatively charged proteins.** T

III. The fluid outside the axon contains a low concentration of K^+ ✓ T

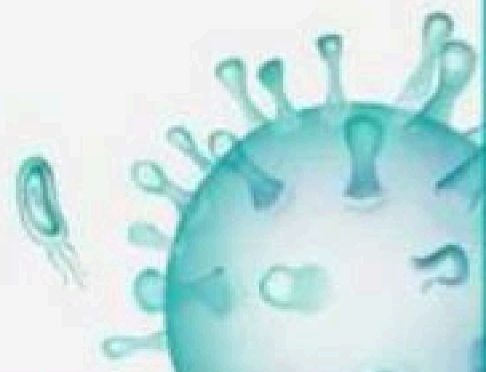
IV. The fluid outside the axon contains a ~~low~~ High. **concentration of** Na^+ X

(1) I and IV are false

(2) II and III are true

(3) 1 and 2

(4) None are correct



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→ negative → +ive.

59. Assertion : In depolarisation process, there is a positive charge on the outer side of the fibre and a negative charge on its inner side. ×

Reason : During depolarisation, permeability of plasma membrane increase for sodium ions, so Na^+ ions move to outer side of the membrane and K^+ enter the nerve fibre. ×

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect. ✓

60. Which part of the human brain controls the urge for eating and drinking?

(1) Forebrain- Hypothalamus

(2) Midbrain- Hypothalamus

(3) Forebrain- Forethalamus

(4) Forebrain- Thalamus

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61. Impulse transmission across an electrical synapse is ____ to that across a chemical synapse

always fast

(1) Always equal

(2) Always slower

(3) Always faster

(4) Sometimes faster

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62. Read the following given statements regarding Na^+ and K^+ identify the correct option

I. Expels $3Na^+$ for every $2K^+$ ions imported

II. Maintains resting potential

III. Works against a concentration gradient

(1) Statement I is correct

(2) Statement II and III are correct

(3) Statement II is correct

(4) Statement I, II and III are correct.

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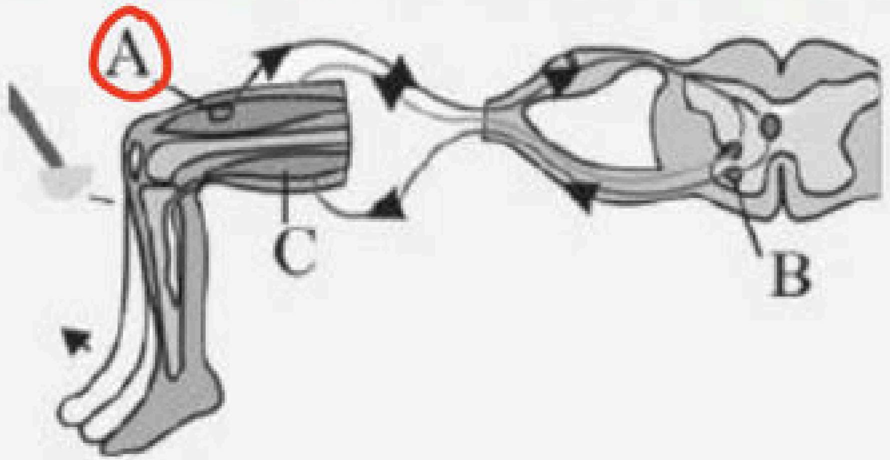
63. Identify the pathway labeled in the given diagram of reflex action?



(1) A - Muscle Spindle , B - Motor neuron, C – Motor endplate

(2) A - Motor neuron, B - Motor endplate, C – Muscle Spindle

(3) A - Motor endplate, B - Muscle Spindle, C – Motor neuron

(4) A - Muscle Spindle, B - Motor endplate, C – Motor neuron

64. For the maintenance of the ionic gradients across the resting membrane, the sodium-potassium pump transports

(1) $3Na^+$ outwards for $2K^+$ into the cell

(2) $2Na^+$ outwards for $2K^+$ into the cell

(3) $3Na^+$ outwards for $3K^+$ into the cell

(4) $2Na^+$ outwards for $3K^+$ into the cell

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65. Statement A : Visceral nervous system is the part of the peripheral nervous system.

Statement B : Visceral nervous system that comprises the whole complex of nerves, fibres, ganglia, and plexuses by which impulses travel from the central nervous system to the viscera and from the viscera to the central nervous system.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

Denchite

66. ____transmit impulse towards cell body and ____ axon transmits impulse away from cell body.

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(1) Dendrites, axon

(2) Dendrites, dendrites

(3) Axon, axon

(4) Axon, dendrites

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67. Schwann cell is absent in

(1) Myelinated neuron

(2) Non myelinated

(3) Astrocytes → blood-brain barrier.

(4) Both (2) and (3)

68. A wave of action potential travelling through the neuron is called as

(1) Resting impulse

(2) Nerve impulse

(3) Motor impulse

(4) Repolarisation

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69. A typical neuron is composed of

(1) Malphigian body and renal tubule

(2) Dendrites and axon

(3) Axon and nucleus

(4) Cell, axon and dendrites

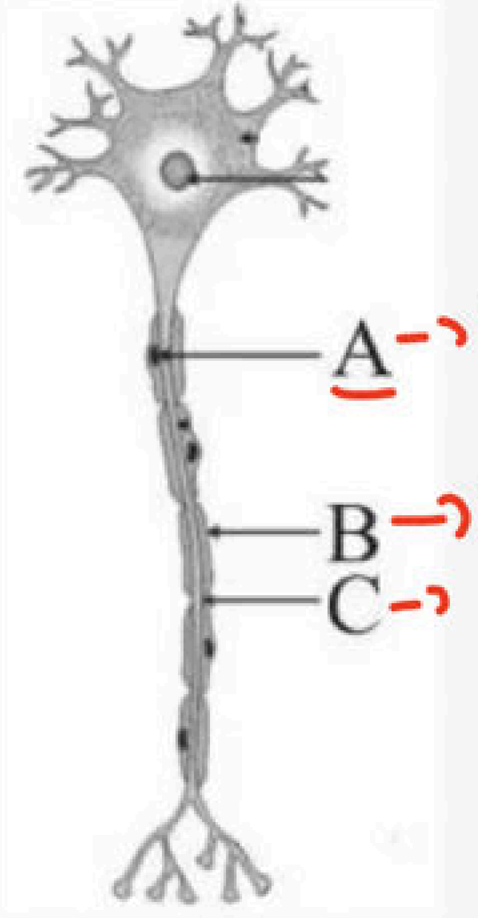
70. Identify the Structures (A-C) around the longest part of neuron?

(1) A - Myelin sheath, B - Node of Ranvier, C – Schwann Cell

(2) A - Schwann cell, B - Myelin sheath, C - Node of Ranvier

(3) A - Myelin Sheath, B - Schwann Cell, C - Node of Ranvier

(4) A - Schwann Cell, B - Node of Ranvier, C – Myelin Sheath



A → S.C
B → M.S
C →

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71. Assertion : The resting membrane of the neuron exhibits polarity of charges.

-ve

Reason : The outer surface of the axonal membrane possesses a negative charge while its inner surface becomes ~~positive~~ly charged.

-ve

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

✓ (3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.

72. The human hind brain comprises three parts, one of which is

(1) Spinal cord

(2) Corpus callosum

(3) Cerebellum

(4) Hypothalamus

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73. In resting stage, the axonal membrane is comparatively more permeable and nearly impermeable to

$\boxed{2025}$

(1) Sodium; potassium

(2) Sodium; calcium

(3) Potassium; sodium

(4) Potassium; calcium

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74. Which of the following is known as the site of information processing and control?

(1) Brain and spinal cord- CNS

(2) Brain and spinal cord-Autonomic nervous system

(3) Brain and spinal cord- PNS

(4) All are correct

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75. Which of the following is false about hindbrain?

(1) Cerebellum has convolutions

(2) Pons, a part of it consist of fibre tracts that interconnects different regions of brain.

(3) Medulla is connected to the spinal cord.

(4) The hind-brain is involved in sensory and motor impulses only

76. Identify the correct statement which describes after events of depolarisation.

i. The rise in the stimulus-induced permeability to Na^+ is extremely shortlived.

ii. It is quickly followed by a rise in ~~impermeability~~ to K^+.

(1) i is correct

(2) ii is correct

(3) Both are correct

(4) Both are incorrect

77. The new potential developed on post-synaptic membrane is

(1) Excitatory always

(2) Inhibitory always

(3) May be excitatory or inhibitory

(4) Neither excitatory nor inhibitory

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78. Coordination is the process through which

(1) Two or more organs interact and complement the functions of one another

(2) Two or more organelles interact and complement the functions of one another

(3) Two or more cells interact and complement the functions of one another

(4) Two or more tissue interact and complement the functions of one another

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79. Four rounded lobes in ~~hindbrain~~ midbrain are :-

(1) Cerebral aqueduct

(2) Corpora quadrigemina = midbrain

(3) Corpora digemina

(4) None are correct

80. What will happen if amygdala of a person is damaged?

(1) Person will forget recent events and cannot commit anything to memory

(2) Person will fail to recognise fearful expression of others / or to express fear in appropriate situation

(3) Person fail to maintain body temperature

(4) Person fails to feel hunger and thirst

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81. Cranial menings from inner layer to outer layer are called

p a d

(1) Piameter, arachnoid, durameter

(2) Durameter, arachnoid, piameter

(3) Arachnoid, durameter, piameter

(4) Arachnoid, piameter, durameter

Title: Neural Control and Coordination

82. Match the parts of the brain given in Column I(Parts of the brain) with their functions given in Column II(Functions). Choose the correct option from the following:

Column I		Column II	
A	Medulla oblongata	P	Body temperature
B	Hypothalamus	Q	Olfaction
C	Cerebral cortex	R	Respiration
D	Limbic system	S	Motor function

(1) $A-P, B-R, C-S, D-Q$

(2) $A-Q, B-S, C-R, D-P$

(3) $A-S, B-P, C-Q, D-R$

(4) $A-R, B-P, C-S, D-Q$

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x   +1   2:32   2025

83. Assertion : Multipolar neurons have two or more axons and one dendrite.

Reason : Bipolar neurons are found usually in the cerebral cortex.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.

84. Read the sentence :

The entire process of response to a PNS stimulation, that occur involuntary i.e., without conscious effort and requires involvement of CNS part is called a reflex action:- Which one is not true about sentence-

(1) Sensory stimuli passes through dorsal surface

(2) All involuntary reflexes are spinal reflex

(3) All spinal reflexes are involuntary X

(4) All are correct

85. Read the given statements regarding mid brain. Identify the correct one. 2025

I. Located between the "thalamus / hypothalamus." and pons of hindbrain

II. Posses a canal named cerebral aqueduct which passes through it.

III. Dorsal part consists of 4 lobes.

(1) I and III

(2) II and III

(3) I and II

(4) All of them.

86. Statement A : The efferent nerve fibres transmit impulses from tissues/ organs to the CNS. X

Statement B : The afferent nerve fibres transmit regulatory impulses from CNS to concerned peripheral tissues/ organs. X

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect. ✓

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87. Durig the propagation of a nerve impulse, the action potential results due to :-

(1) Rapid influx of Na^{+2}

(2) Rapid influx of Na^{+}

(3) Rapid influx of K^{+}

(4) Rapid influx of K^{+2}

88. Statement A : During impulse, the rise in the stimulusinduced permeability to is extremely Na^+ short lived.

Statement B : It is quickly followed by a rise in permeability to K^+.

NEET 2025

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

Na^+

(3) Both statements are correct.

(4) Both Statements are incorrect.

89. Association areas of the brain are

(1) Always clearly sensory areas

(2) Always clearly motor areas

(3) Neither clearly sensory nor motor areas

(4) All are correct

90. Multipolar neurons are found in the

(1) Retina of eye

(2) Cerebral cortex

(3) Embryonic stage

(4) None of these

91. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Mid – Brain	P	A canal passess through the midbrain.
B	Cerebral aqueduct	Q	Midbrain consists mainly of four round lobes
C	Corpora quadrigemina	R	Located between thalamus & hypothalamus of forebrain

(1) A-Q, B-R, C-P (2) A-Q, B-P, C-R

(3) A-R, B-P, C-Q (4) A-R, B-Q, C-P

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92. Match Column I with Column II :

Choose the correct answer from the options given below :

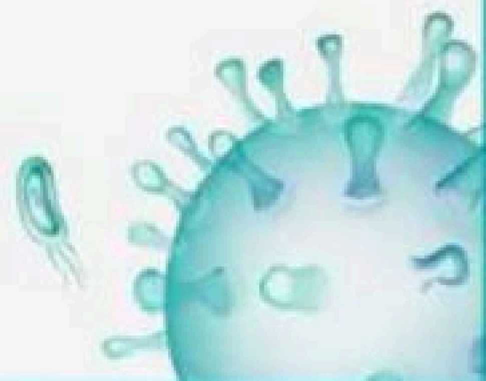
Column I		Column II	
A	Pons	P	Provides additional space for neurons, regulates posture and balance
B	Hypothalamus	Q	Controls respiration and gastric secretions
C	Medulla	R	Connects different regions of the brain
D	Cerebellum	S	Neuro secretory cells

(1) A – Q, B – R, C – P, D – S

(2) A – R, B – S, C – Q, D – P

(3) A – P, B – R, C – Q, D – S

(4) A – Q, B – P, C – R, D – S



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93. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Cranial meninges	P	Outer layer which is in contact with brain tissue
B	Dura mater	Q	Inner layer which is in contact with brain tissue
C	Arachnoid	R	Brain is covered by membrane inside the skull
D	Pia mater	S	A very thin middle layer

(1) A-Q, B-P, C-Q, D-R

(2) A-Q, B-P, C-R, D-P

(3) A-P, B-R, C-Q, D-R

(4) A-R, B-P, C-S, D-Q

94. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Cerebral hemisphere	P	Hemispheres are connected by a tract of nerve fibres
B	Corpus callosum	Q	Cerebrum longitudinally divides into two halves
C	Cerebral cortex	R	Regions responsible for memory & communication
D	Associated area	S	Contains motor areas, sensory areas & large regions that are neither sensory nor motor in function

(1) A-P, B-Q, C-S, D-R

(2) A-Q, B-P, C-S, D-R

(3) A-S, B-R, C-Q, D-P

(4) A-S, B-R, C-P, D-Q

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95. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Synapse	P	The membranes of the pre- and post- synaptic neurons are separated by fluid filled space
B	Synaptic cleft	Q	Junction through which nerve impulse is transmitted from one neuron to another
C	Axon Terminal	R	Chemicals involved in the transmission of impulses at synapse
D	Neurotransmitters	S	Contain vesicles filled with acetylcholine

(1) A-P, B-R, C-Q, D-S
(2) A-S, B-Q, C-R, D-P
(3) A-P, B-Q, C-R, D-S
(4) A-Q, B-P, C-S, D-R

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96. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Brain stem	P	Contains centre which control respiration, cardiovascular reflexes & gastric secretions
B	Pons	Q	Consists of fibre tracts that interconnect different regions of the brain
C	Cerebellum	R	Convoluted surface in order to provide the additional space for many more neurons
D	Medulla	S	Formed by midbrain & hindbrain

(1) A-S, B-Q, C-R, D-P (2) A-R, B-P, C-Q, D-S

(3) A-R, B-Q, C-P, D-S (4) A-P, B-Q, C-R, D-S

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97. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Thalamus	P	Control body temperature, urge for eating & drinking
B	Hypothalamus	Q	Involved in regulation of sexual behavior, expression of emotional reactions
C	Cerebrum	R	Major coordinating centre for sensory & motor signaling
D	Limbic system	S	Forms the major part of human brain

(1) A-P, B-Q, C-R, D-S (2) A-P, B-R, C-S, D-Q

(3) A-R, B-P, C-S, D-Q (4) A-Q, B-R, C-S, D-P

98. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Stimulus	P	Muscle spindle
B	Receptor	Q	Motor end plate
C	Effector	R	Extremely hot, cold objects
D	Reflex action	S	Knee jerk reflex

(1) A-Q, B-R, C-S, D-P

(2) A-Q, B-R, C-P, D-S

(3) A-R, B-Q, C-S, D-P

(4) A-R, B-P, C-Q, D-S

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99. Statement A : A canal called the corpora quadrigemina passes through the mid brain.

Statement B : The dorsal portion of the midbrain consists mainly of four round lobes called cerebral aqueduct.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

100. Statement A : At a chemical synapse, the membranes of the pre- and post-synaptic neurons are separated by a fluid filled space called ~~synaptic knob~~. X

Statement B : A nerve impulse is transmitted from one neuron to another through junctions called synapses. ✓

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct. ✓

(3) Both statements are correct.

(4) Non

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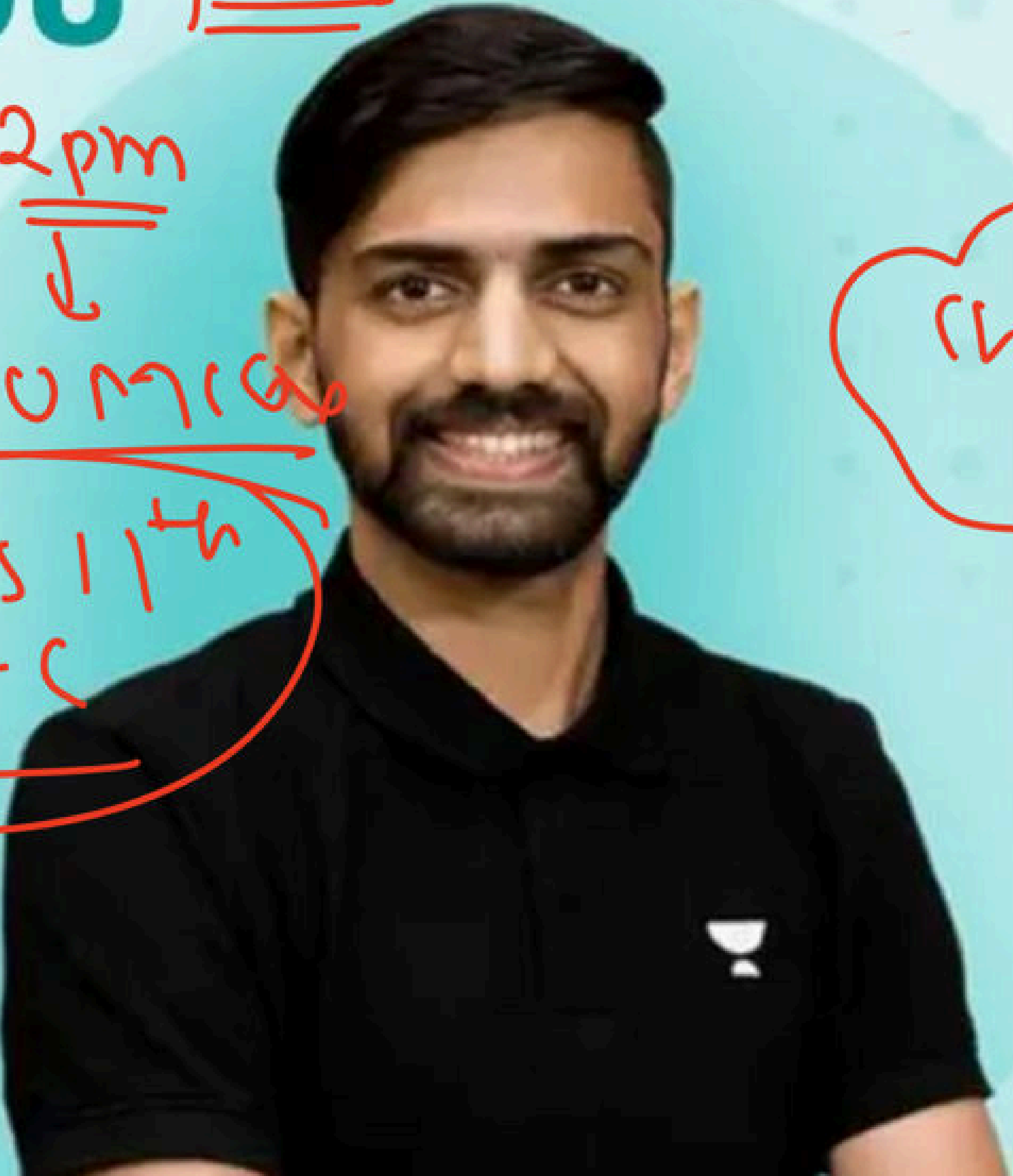
THANK YOU

8 PM

29th – 10am, 2pm
↓
100 mcqs

Class 11th
M.I.C

BIOLOGYLIVE



Chemical control and Inh
↓
100 MCQs